Thermodynamic description of the Cu–Zn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

Synthetic document generated by the ORCHID pipeline benchmark; not a real publication. CC BY 4.0. Derived from the SGTE binary dataset collection (B. Hallstedt, Calphad 89 (2025) 102833, CC BY 4.0) and the original assessment cited in the dataset.

Abstract

This synthetic document describes the Cu–Zn binary system as calculated with pycalphad from the SGTE dataset CuZn-08Gie.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, (Cu) + L → β, lies at 899 °C.

1. Introduction

The Cu–Zn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as peritectic, intermetallic rich, order disorder. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of W. Gierlotka, S.-W. Chen, J. Mater. Res., 23, 258-63(2008). The calculations use pycalphad 0.11.2 at 1 atm.

The melting point of Cu reproduced by the dataset is 1085 °C. The melting point of Zn reproduced by the dataset is 420 °C.

2. Phase diagram

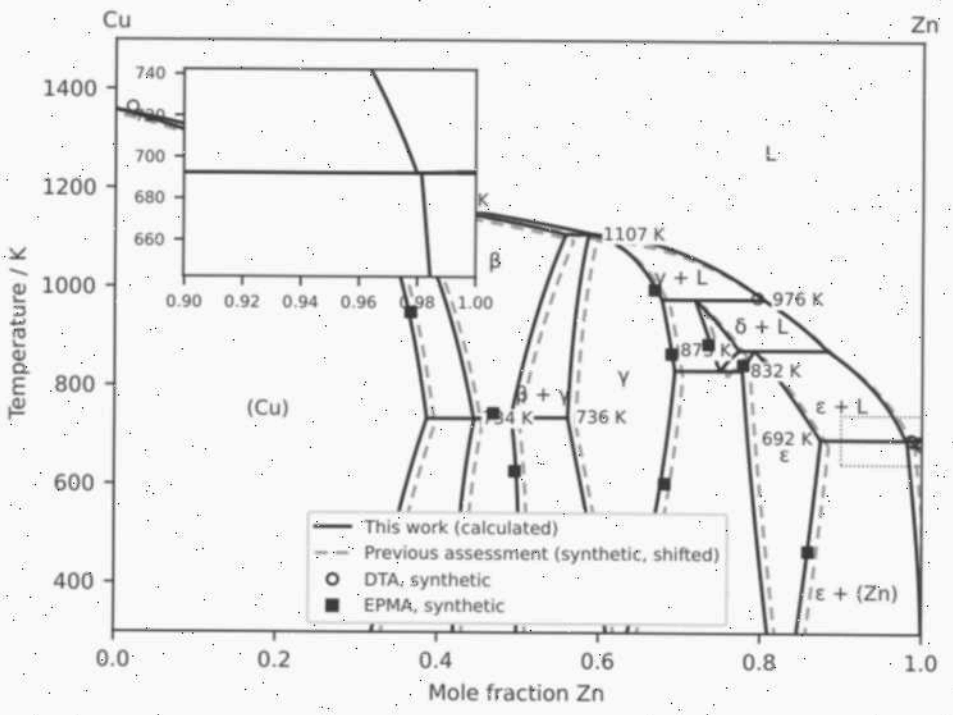


Fig. 1. Calculated Cu–Zn phase diagram with synthetic experimental-style points and a dashed, deliberately shifted second assessment (inset: enlarged view near the lowest invariant).

We obtain 899 °C for the peritectic reaction ((Cu) + L → β), where L has a mole fraction of 0.375. Peritectic equilibrium β + L → γ is found at 834 °C and x(Zn) = 0.593 (L). The peritectic reaction γ + L → δ is calculated at 703 °C, with L at x(Zn) = 0.804. The dataset places a

peritectic (δ + L → ε) at 600 °C; the composition of L is a mole fraction of 0.883. We obtain 559 °C for the eutectoid reaction (δ → γ + ε), where δ has a mole fraction of 0.749. The dataset places a peritectoid (β + γ → β′) at 463 °C; the composition of β′ is a mole fraction of 0.491. The eutectoid reaction β → (Cu) + β′ is calculated at 460 °C, with β at x(Zn) = 0.445. The dataset places a eutectic (L → ε + (Zn)) at 419 °C; the composition of L is a mole fraction of 0.980.

Table 1. Invariant reactions in the Cu–Zn system (calculated).

Reaction	Type	T (°C)	x(Zn), 1	x(Zn), 2	x(Zn), 3
(Cu) + L → β	peritectic	899.4	0.319	0.375	0.354
β + L → γ	peritectic	834.2	0.558	0.593	0.586
γ + L → δ	peritectic	703.1	0.677	0.804	0.718
δ + L → ε	peritectic	600.2	0.773	0.883	0.792
δ → γ + ε	eutectoid	558.8	0.749	0.693	0.777
β + γ → β′	peritectoid	462.7	0.491	0.561	0.491
β → (Cu) + β′	eutectoid	460.4	0.445	0.387	0.445
L → ε + (Zn)	eutectic	419.1	0.980	0.874	0.981

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Thermodynamic properties

At 1450 K the integral mixing enthalpy of the liquid has its minimum, −6.44 kJ/mol, at x(Zn) = 0.440. The calculated enthalpy of formation of γ at 298.15 K is −10.64 kJ/mol per mole of atoms.

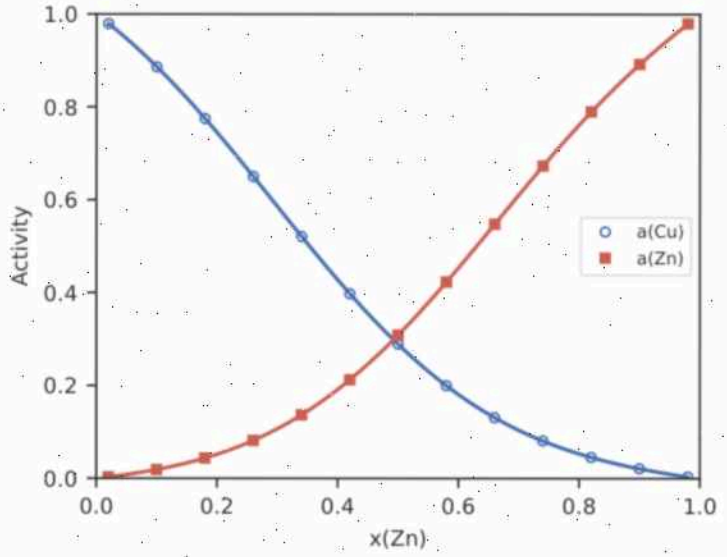


Fig. 2. Calculated activities in the liquid at 1450 K (reference: pure liquids).

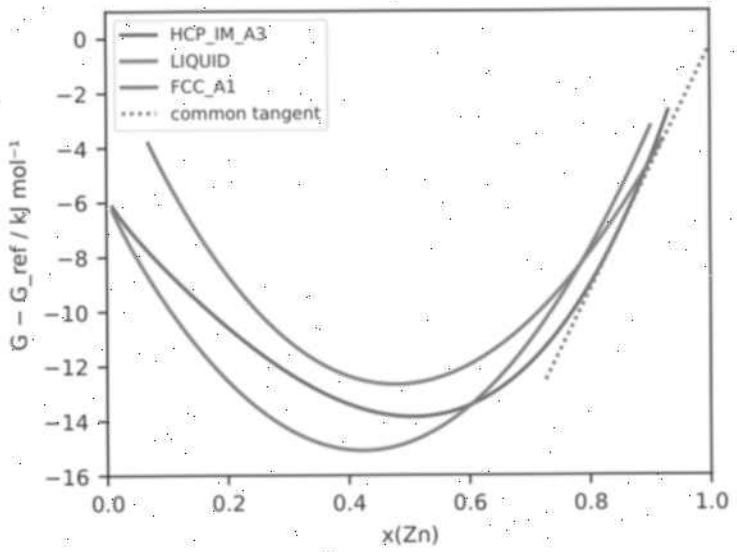


Fig. 3. Calculated Gibbs energy curves at 760 K.

Table 2. Calculated formation properties of the intermediate phases (Cu–Zn).

Phase	at.% Zn	ΔfH(298 K) / kJ mol⁻¹	S(298 K) / J mol⁻¹ K⁻¹
β	49.7	-9.43	41.75
δ	75.0	-5.76	44.74
γ	64.0	-10.64	41.14
β′	46.0	-10.41	39.79
ε	83.0	-4.89	43.61

Reference: stable elements (SER) at 298.15 K.

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.